## Slope and Rate of Change — Diagnosing Inverted and Sign-Reversed Calculations

---

Slope is **rise over run**, and both differences must be taken in the **same order**:

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

- Flipping the fraction gives the **inverted** slope (run over rise).
- Starting with point 1 on top and point 2 on the bottom gives the **sign-reversed** slope.
- In each problem, decide which of those two errors was made (if any), then compute the correct value. Show the subtraction you used.

**1.** Warm up. Find the slope of the line through $(2, 3)$ and $(6, 11)$. Label your rise and your run before you divide.

Answer: ________________

**2.** **Find and fix.** Devi finds the slope through $(-3, 5)$ and $(2, -5)$. She writes:

$$m = \frac{5 - (-5)}{2 - (-3)} = \frac{10}{5} = 2$$

Devi's arithmetic is correct, but her answer is not. Name the error in her set-up, then give the correct slope.

Answer: ________________

**3.** **Find and fix.** Mika finds the slope through $(1, 4)$ and $(5, 6)$. He writes:

$$m = \frac{5-1}{6-4} = \frac{4}{2} = 2$$

Name Mika's error, then give the correct slope as a fraction.

Answer: ________________

**4.** A tank is draining. The table shows the water level, in litres, at four times.

| **Hours** | 0 | 2 | 4 | 6 |
|---|---|---|---|---|
| **Litres** | 46 | 40 | 34 | 28 |

Find the **change** in the water level from hour 0 to hour 6. A change that goes down is negative, so write your answer with its sign.

Answer: ________________

**5.** Use the same draining tank.

| **Hours** | 0 | 2 | 4 | 6 |
|---|---|---|---|---|
| **Litres** | 46 | 40 | 34 | 28 |

Find the rate of change of the water level, in litres per hour, between hour 0 and hour 6. Keep the level difference on top and the time difference underneath, both taken in the same order.

Answer: __________________

**6.** **Find and fix.** Ty finds the slope through $(5, 9)$ and $(1, 1)$. He writes:

$$m = \frac{1-9}{5-1} = \frac{-8}{4} = -2$$

Both of Ty's differences came out of the same two points, so his arithmetic looks fine — but he mixed the order. Name the error and give the correct slope. Then say whether the line rises or falls from left to right.

Answer: __________________

**7.** A candle burns at a steady rate. At 3 minutes it is 18 cm tall, and at 11 minutes it is 6 cm tall.
Jo reports the rate of change as $\frac{11-3}{6-18} \approx -0.67$ cm per minute. Explain what Jo inverted, then find the correct rate of change of the candle's height, in centimetres per minute.

Answer: ________________

**8. Challenge.** A line passes through $(a, 4)$ and $(7, -2)$, and its slope is $-3$. Priya sets up

$$\frac{7-a}{-2-4} = -3$$

and reports $a = -11$.
Diagnose Priya's set-up error, write the correct equation, and solve it for $a$.

Answer: ________________